

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

SRI VENKATESWARA UNIVERSITY COLLEGE OF ENGINEERING:

TIRUPATI-517502

R-23– Scheme of Instructions effective from the academic year 2023 – 2024

Programme Scheme



**SRI VENKATESWARA UNIVERSITY COLLEGE OF ENGINEERING
(AUTONOMOUS)**

SRI VENKATESWARA UNIVERSITY

TIRUPATI-517502 (A.P), INDIA.

SRI VENKATESWARA UNIVERSITY COLLEGE OF ENGINEERING: TIRUPATI – 517 502
Department of Electronics and Communication Engineering-Scheme of Instruction- (CBCS) effective from the Academic Year 2023-2024

M.Tech (PG) (Electronics and Communication Engineering)
Specialization: Communication Systems (CM)

I Semester

S.No	Category	Course Code	Course Title	Scheme of Instruction (Hours/Week)				No. of Credits	Scheme of Evaluation		
				Lecture	Tutorial	Practical	Total		Sessional Marks	Semester End Examination Marks	Total
1	PCC	CM11C	Advanced Digital Signal Processing (Common to Signal Processing SP11C)	3	1	0	4	4	40	60	100
2	PCC	CM12C	Wireless and Mobile Communication	3	1	0	4	4	40	60	100
3	PCC	CM13C	DSP Architecture (Common to Signal Processing SP13C)	3	1	0	4	4	40	60	100
4	PEC	CM14C	Programme Elective- I	3	0	0	3	3	40	60	100
5	PEC	CM15C	Programme Elective- II	3	0	0	3	3	40	60	100
6	PCC	CM16L	Advanced Digital Signal Processing Lab (Common to Signal Processing SP16L)	0	0	3	3	1.5	40	60	100
7	PCC	CM17L	Wireless and Mobile Communication Lab	0	0	3	3	1.5	40	60	100
8	MAC	CM18C	Research Methodology and IPR	3	0	0	3	3	40	60	100
Total				18	3	6	27	24	320	480	800

List of Programme Elective Course I(CM14C)	List of Programme Elective Course II(CM15C)
1. Optical Networks	1. Cognitive Radio
2. Statistical Information Processing	2. Voice and Data Networks (Common to signal Processing SP15C)
3. Wireless Sensor Networks (Common to signal Processing SP14C)	3. Secured Communication
4. High Performance Networks	4. Electromagnetic Interference and Compatibility

II Semester

S.No.	Category	Course Code	Course Title	Scheme of Instruction (Hours/Week)				No. of Credits	Scheme of Evaluation		
				Lecture	Tutorial	Practical	Total		Sessional Marks	Semester End Examination Marks	Total
1	PCC	CM21C	Antennas and Radiating Systems	3	1	0	4	4	40	60	100
2	PCC	CM22C	Advanced Communication Networks	3	1	0	4	4	40	60	100
3	PEC	CM23C	Programme Elective Course - III	3	0	0	3	3	40	60	100
4	PEC	CM24C	Programme Elective Course - IV	3	0	0	3	3	40	60	100
5	PCC	CM25L	Antennas and Radiating Systems Lab	0	0	3	3	1.5	40	60	100
6	PCC	CM26L	Advanced Communication Networks Lab	0	0	3	3	1.5	40	60	100
7	VAC	CM27C	Cyber Security	2	0	2	4	3	100	-	100
8	PCC	CM28M	Mini Project with Seminar	0	0	4	4	2	100	-	100
Total				14	2	12	28	22	440	360	800

List of Programme Elective Courses III(CM23C)	List of Programme Elective Courses IV(CM24C)
1. Satellite Communication	1. Random processes and Queueing models. (Common to Signal Processing SP24C)
2. IoT and Applications (Common to signal Processing SP23C)	2. Pattern Recognition and Machine Learning (Common to Signal Processing SP21C)
3. RF and Microwave Circuit Design	3. Programmable Networks-SDN, NFV
4. MIMO Systems	4. Remote Sensing (Common to Signal Processing SP24C)

III Semester

S.No.	Category	Course Code	Course Title	Scheme of Instruction (Hours/Week)				No. of Credits	Scheme of Evaluation		
				Lecture	Tutorial	Practical	Total		Sessional Marks	Semester End Examination Marks	Total
1	OEC	CM31C	Open Elective Course (Through MOOCS)	0	0	0	0	3	100	-	100
2	PCC	CM32I	Industrial/ Research Internship (Min of 4 Weeks)	0	0	0	0	3	100	-	100
3	PCC	CM33D	Dissertation Work Phase-I	0	0	24	24	12	40	60	100
Total				0	0	20	20	18	240	60	300

IV Semester

S.No.	Category	Course Code	Course Title	Scheme of Instruction (Hours/Week)				No. of Credits	Scheme of Evaluation		
				Lecture	Tutorial	Practical	Total		Sessional Marks	Semester End Examination Marks	Total
1	PCC	CM41D	Dissertation Work Phase-II and Viva-Voce	0	0	20	20	10 + 06 = 16	40	60	100
Total				0	0	20	20	16	40	60	100

M. Tech. (Electronics & Communication Engineering)

Curriculum Structure

Specialization: Communication systems (CM)

Vision of The Institute

The Vision of Sri Venkateswara University College of Engineering is to be the leader in the creation and development of globally competitive human capital in Engineering Education for Technological, Economical and Social Enrichment of the Society, through its open and flexible Academic Programs.

Mission of The Institute

- To be recognized as a premier institution offering Engineering Education programs, training human resources oriented to problem solving and system development.
- To carry out research in Engineering and Technology relevant to all segments of society.
- To assume leadership in sustainable technological growth of the Indian society.
- To be a natural destination for excellence and diversity in thought and practice.

Department of Electronics & Communication Engineering

Vision

To be a lead department imparting quality and value embedded higher education and research emphasizing freedom of learning and practice.

Mission

1. Transforming students into full-fledged professionals and to become leaders in dynamic global environment.
2. Augmenting knowledge and technologies in rapidly advancing fields of Electronics and Communication Engineering.
3. Promoting in depth research and create Centre of excellence in thrust areas.

Program Outcomes (POs) for PG programs

POs are statements that describe what students are expected to know and be able to do upon graduating from the program. These relate to the skills, knowledge, analytical ability attitude and behaviour that students acquire through the program.

The POs essentially indicate what the students can do from subject-wise knowledge acquired by them during the program. As such, POs define the professional profile of a graduate of PG Engineering Program.

NBA has defined the following three POs for a graduate of PG Engineering Program:

1. **PO1:** An ability to independently carry out research/investigation and development work to solve practical problems.
2. **PO2:** An ability to write and present a substantial technical report/document.
3. **PO3:** Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.

Program Specific Outcomes (PSOs):

M. Tech. in Communication Systems graduates will be able to:

1. **PSO 1.** Integrate knowledge to identify, formulate, solve complex problems and meet the challenges in the domain of communication and networking.
2. **PSO 2.** Use the concepts of communications, networking and signal processing to simulate algorithms in virtual platforms and implement them in real time environment.

Program Educational Objectives (PEOs):

M. Tech. in Communication Systems Program, graduates will be able to:

- . **PEO 1.** Apply the concepts of mathematics and engineering principles in the domain of communication and networking and analyze advantages and disadvantages of using the various communication technologies.
- . **PEO 2.** Solve numerical problems and write algorithms using communication and networking platforms and different software tools to evolve viable communication systems.
- . **PEO 3.** Inculcate professional ethics, integrity and social responsibility, enhance communication skills, and practice effective teamwork to pursue research or attain a career in the domain of Communication Engineering.

CM11C: Advanced Digital Signal Processing (Common to signal processing)

Instruction Hours/week: (L-T-P-C): 3-1-0-4

Credits : 4

Sessional Marks : 40

Semester-End Examination : 60

Course Description: This course will provide rigorous foundations in Digital filter structures, Multirate signal processing, Linear prediction, Adaptive filters and Spectral estimation. This course emphasizes the use of digital signal processing techniques for designing digital systems used in Communications, Control, Media Applications etc.

Prerequisites: Signals and Systems, Digital Signal Processing.

Course Objectives: To enable the student to

1. Understand to develop FIR & IIR filter structures depending upon the given applications.
2. Understand theory of Multirate DSP, solve numerical problems and write algorithms.
3. Understand theory of prediction and solution of normal equations.
4. Understand the different types of adaptive filters used in signal processing applications.
5. Discuss different methods of spectrum estimation and analysis.

Course Outcomes: After completion of the course, student will be able to

1. Learn and implement the digital filters to computational complexity problems.
2. Analyze multirate DSP systems and Design a decimator and integrator including multi-stages.
3. Apply theory of prediction and find solution of normal equations
4. Design different types of filters such as Adaptive filters, poly phase filters, Wiener filters, ARMA lattice-ladder filters, etc.
5. Estimate the power spectrum by using different methods.

Contents:

Unit 1

Digital filter structures: Basic FIR/IIR filter structures, FIR/IIR Cascaded lattice structures, Parallel structures, all pass realization of IIR transfer functions, Sine- cosine generator, Computational complexity of filter structures.

Unit 2

Multirate Digital Signal Processing: Decimators and Interpolators, Sampling rate conversion, multistage decimator & interpolator, poly phase filters, Quadrature mirror filter (QMF) banks, Conditions for perfect reconstruction, digital filter banks, Applications in sub-band coding.

Unit 3

Linear Prediction and Optimum Linear Filters: Innovations Representation of a Stationary Random Process, Forward and Backward linear prediction, Solution of the Normal Equations, Properties of linear prediction-Error Filter, AR Lattice and ARMA Lattice-Ladder Filters, Wiener Filters for Filtering and Prediction.

Unit 4

Adaptive Filters, Applications, Gradient Adaptive Lattice, Minimum mean square criterion, The Least-Mean-Square (LMS) algorithm, Normalized LMS (NLMS), The Leaky LMS, Recursive Least Square (RLS) algorithm.

Unit 5

Power Spectral Estimation: Estimation of Spectra from Finite Duration Observations of a Signal, Periodogram, Use DFT in power Spectral Estimation, Bartlett, Welch and Blackman, Tukey methods, Comparison of performance of Non-Parametric Power Spectrum Estimation Methods.

Parametric Methods of Power Spectrum Estimation: Relationship between Auto-Correlation and Model Parameters, Yule-Walker, Burg and Unconstrained Least Squares Methods, Moving Average (MA) and ARMA Models Minimum Variance Method, Eigen analysis Algorithms for Spectrum Estimation.

Text Books:

1. **John G. Proakis and Dimitris G. Manolakis**, Digital Signal Processing: Principles, Algorithms and Applications, 5th Edition, **ISBN-13: 9780137348657, Published** by Pearson, 2021.
2. John G. Proakis, Digital Signal Processing Principles, Algorithms, and Applications, Prentice Hall International Inc, 4th Edition, 2012.
3. Sanjit Kumar Mitra, Digital Signal Processing- A Computer Based Approach, 4th edition, McGraw Hill Education, 2011.
4. D.G. Manolakis, V.K. Ingle and S.M.Kogon, "Statistical and Adaptive Signal Processing", McGraw Hill, 2000.

References:

1. N. J. Flieg Multirate Digital Signal Processing: Multirate Systems - Filter Banks – Wavelets, 1st Edition, John Wiley and Sons Ltd, 1999.

2. Sanjit Kumar Mitra, and Yonghong Kuo. Digital Signal Processing: A Computer-Based Approach, 4 th. Edition McGraw-Hill Higher Education, New York, 2013.
3. M. H. Hayes, “Statistical Digital Signal Processing and Modeling”, John Wiley & Sons Inc., 2002.
4. Simon Haykin, “Adaptive Filter Theory”, 4th Edition, Prentice Hall, 2001.

CM12C: Wireless and Mobile Communication

Instruction Hours/week : (L-T-P-C): 3-1-0-4

Credits : 4

Sessional Marks

: 40

Semester-End Examination : 60

Course Description: This course provides a comprehensive overview and advanced knowledge of modern mobile and wireless communication systems. Building on the prior knowledge on digital communications, students develop further understanding on the challenges and opportunities brought by the wireless medium in designing current and future wireless communication systems and networks

Course Objectives:

1. To provide the students with an understanding of the cellular concepts, frequency reuse, handoff strategies.
2. To enable the students to analyze and understand wireless and mobile cellular communication systems over stochastic fading channels.
3. To provide the students with an understanding of Co-channel and Non-Co channel Interference.
4. To give students an understanding of cell coverage for signal and traffic diversity techniques and mobile antennas.
5. To give the students an understanding of frequency management, channel assignment and types of handoff.

Contents

UNIT 1

Cellular Communication Fundamentals: Cellular system design, Frequency reuse, cell splitting, handover concepts, Co channel and adjacent channel interference, interference reduction techniques and methods to improve cell coverage, frequency management and channel assignment. GSM architecture and interfaces, GSM architecture details, GSM subsystems, GSM Logical Channels, Data Encryption in GSM, GPRS, IS 95, Forward Link and Reverse links.

UNIT 2

Spectral efficiency analysis based on calculations for Multiple access technologies: TDMA, FDMA and CDMA, Comparison of these technologies based on their signal separation techniques, advantages, disadvantages and application areas. Wireless network planning (Link budget and power spectrum calculations)

UNIT 3

Mobile Radio Propagation: Large Scale Path Loss, Free Space Propagation Model, Reflection, Ground Reflection (Two-Ray) Model, Diffraction, Scattering, Practical Link Budget Design using Path Loss Models, Outdoor Propagation Models, Indoor Propagation Models, Signal Penetration into Buildings. Small Scale Fading and Multipath Propagation, Impulse Response Model, Multipath Measurements, Parameters of Multipath channels, Types of Small-Scale Fading: Time Delay Spread; Flat, Frequency selective, Doppler Spread; Fast and Slow fading.

UNIT 4

Equalization, Diversity: Equalizers in a communications receiver, Algorithms for adaptive equalization, diversity techniques, space, polarization, frequency diversity, Interleaving.

UNIT 5

5G Architecture Add Network, 5G Radio, Access Technologies, Security for 5G Communications, 5G challenges and future scope.

Course Outcomes: Upon completion of the course, students will be able to

1. Design appropriate mobile communication systems by applying frequency-reuse concept, and analyze its effects on interference, system capacity, handoff techniques.
2. Distinguish various multiple-access techniques e.g. FDMA, TDMA, CDMA, and their advantages and disadvantages.
3. Analyze path loss and interference for wireless telephony and their influences on a mobile communication system's performance.
4. Analyze and design CDMA system with the knowledge of forward and reverse channel details, advantages and disadvantages of it.
5. Understand technologies like 2G, 3G, 4G, 5G, etc.

TEXT BOOKS:

1. T.S.Rappaport, "Wireless Communications Principles and Practice", 2nd edition, PHI, 2002.
2. William C.Y.Lee, "Mobile Cellular Telecommunications Analog and Digital Systems", 2nd edition, TMH, 1995.
3. Ajit Singh, "5G Simplified", Shroff Publishers, 2019.

REFERENCES:

1. V.K.Garg, J.E.Wilkes, "Principle and Application of GSM", Pearson Education, 5th edition, 2008.
2. V.K.Garg, "IS-95 CDMA & CDMA 2000", Pearson Education, 4th edition, 2009.
3. Asha Mehrotra, "A GSM system Engineering" Artech House Publishers Boston, London, 1997.

CM13C: DSP Architecture (Common To Signal Processing)

Instruction Hours/week : (L-T-P-C): 3-1-0-4

Credits : 4

Sessional Marks

: 40

Semester-End Examination : 60

Course Description: This course explains fundamental concepts of Digital Signal Processing and implementation of various applications on Advanced Processor. Helps students to understand architecture of advanced Digital Signal Processor and how to program it for signal processing applications.

Course Objectives:

1. To describe features and architectural improvements of DSP processors.
2. Introduce addressing modes and instruction description of TMS320C6x processors.
3. To demonstrate data representation in DSP Processors and FIR filters.
4. To demonstrate the usefulness of the adaptive filters and learn techniques of code optimization.

Course Outcomes:

1. At the end of this course, students will be able to
2. Identify and formalize architectural level characterization of P-DSP hardware
3. Ability to design, programming (assembly and C), and testing code using Code Composer Studio environment
4. Deployment of DSP hardware for Control, Audio and Video Signal processing applications
5. Understanding of major areas and challenges in DSP based embedded systems

Syllabus Contents:

Unit 1

Programmable DSP Hardware: Processing Architectures (von Neumann, Harvard), DSP core algorithms (FIR, IIR, Convolution, Correlation, FFT), IEEE standard for Fixed and Floating Point Computations, Special

Architectures Modules used in Digital Signal Processors (like MAC unit, Barrel shifters), On-Chip peripherals, DSP benchmarking.

Unit 2

Structural and Architectural Considerations: Parallelism in DSP processing, Texas Instruments.TMS320 Digital Signal Processor Families, Fixed Point TI DSP Processors: TMS320C1X and TMS320C2X Family,TMS320C25 – Internal Architecture, Arithmetic and Logic Unit, Auxiliary Registers, Addressing Modes (Immediate, Direct and Indirect, Bit-reverse Addressing), Basics of TMS320C54x and C55x Families in respect of Architecture improvements and new applications fields, TMS320C5416 DSP Architecture, Memory Map, Interrupt System, Peripheral Devices, Illustrative Examples for assembly coding.

Unit 3

VLIW Architecture: Current DSP Architectures, GPUs as an alternative to DSP Processors, TMS320C6X Family, Addressing Modes, Replacement of MAC unit by ILP, Detailed study of ISA, Assembly Language Programming, Code Composer Studio, Mixed C and Assembly Language programming, On-chip peripherals, Simple applications developments as an embedded environment.

Unit 4

Multi-core DSPs: Introduction to Multi-core computing and applicability for DSP hardware, Concept of threads, introduction to P-thread, mutex and similar concepts, heterogeneous and homogenous multi-core systems, Shared Memory parallel programming –OpenMP approach of parallel programming, PRAGMA directives, OpenMP Constructs for work sharing like for loop, sections, TI TMS320C6678 (Eight Core subsystem).

Unit 5

FPGA based DSP Systems: Limitations of P-DSPs, Requirements of Signal processing for Cognitive Radio (SDR), FPGA based signal processing design-case study of a complete design of DSP processor,High Performance Computing using P-DSP: Preliminaries of HPC, MPI, OpenMP, multicore DSP as HPC infrastructure.

References:

1. M. Sasikumar, D. Shikhare, Ravi Prakash, “Introduction to Parallel Processing”, 1st Edition, PHI, 2006.
2. Fayez Gebali, “Algorithms and Parallel Computing”,1st Edition, John Wiley & Sons, 2011
3. Rohit Chandra, Ramesh Menon, Leo Dagum, David Kohr, DrorMaydan,

- Jeff McDonald, "Parallel Programming in OpenMP", 1st Edition, Morgan Kaufman, 2000.
4. Ann Melnichuk, Long Talk, "Multicore Embedded systems", 1st Edition, CRC Press, 2010.
 5. Wayne Wolf, "High Performance Embedded Computing: Architectures, Applications and Methodologies", 1st Edition, Morgan Kaufman, 2006.
 6. E.S.Gopi, "Algorithmic Collections for Digital Signal Processing Applications Using MATLAB", 1st Edition, Springer Netherlands, 2007.

CM14C: Program Elective-I

1. Optical Networks

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination : 60

Course Description: Optical fibers have several advantages over conventional copper cables. Optical communication enjoys high bandwidth and thus useful for transmitting huge information securely and with less repeaters to longer distances. The performance of optical system is superior to conventional communication and thus has widespread use in industry.

Course Objectives: Optical Communication has got greater applicability and students of Communication discipline should master the subject. This course will enable the students:

1. To understand deeply the fundamental aspects of optical communication
2. To analyze the types of fibers and understand the performance by comparison
3. To design optical receivers of optical systems
4. To simulate the performance through parameters like BER and S/N Ratio
5. To apply the principles of optical communication to networks.

Course Outcomes:

At the end of this course, students will be able to

1. Contribute in the areas of optical network and WDM network design.
2. Implement simple optical network and understand further technology developments for future enhanced network.

Syllabus Contents:

Unit 1

SONET/SDH: optical transport network, IP, routing and forwarding, multiprotocol label switching.

WDM network elements: optical line terminals and amplifiers, optical add/drop multiplexers, OADM architectures, reconfigurable OADM, optical cross connects.

Unit 2

Control and management: network management functions, optical layer services and interfacing, performance and fault management, configuration management, optical safety.

Unit 3

Network Survivability: protection in SONET/SDH & client layer, optical layer protection schemes

Unit 4

WDM network design: LTD and RWA problems, dimensioning wavelength routing networks, statistical dimensioning models.

Unit 5

Access networks: Optical time division multiplexing, synchronization, header processing, buffering, burst switching, test beds, Introduction to PON, GPON, AON.

References:

1. Rajiv Ramaswami, Sivarajan, Sasaki, "Optical Networks: A Practical Perspective", MK, Elsevier, 3rd edition, 2010.
2. C. Siva Ram Murthy and Mohan Gurusamy, "WDM Optical Networks: Concepts Design, and Algorithms", PHI, EEE, 2001.

CM14C: Program Elective-I

2. Statistical Information Processing

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination : 60

Course Outcomes:

At the end of this course, students will be able to

1. Characterize and apply probabilistic techniques in modern decision systems, such as information systems, receivers, filtering and statistical operations.
2. Demonstrate mathematical modelling and problem solving using such models.
3. Comparatively evolve key results developed in this course for applications to signal processing, communications systems.
4. Develop frameworks based in probabilistic and stochastic themes for modelling and analysis of various systems involving functionalities in decision making, statistical inference, estimation and detection.

Syllabus Contents:

Unit 1

Review of random variables: Probability Concepts, distribution and density functions, moments, independent, uncorrelated and orthogonal random variables; Vector-space representation of Random variables, Vector quantization, Tchebay chef inequality theorem, Central Limit theorem, Discrete & Continuous Random Variables.

Random process: Expectations, Moments, Ergodicity, Discrete-Time Random Processes Stationary process, autocorrelation and auto covariance functions, Spectral representation of random signals, Properties of power spectral density, Gaussian Process and White noise process.

Unit 2

Random signal modelling: MA(q), AR(p), ARMA(p,q) models, Hidden Markov Model & its applications, Linear System with random input, Forward and Backward Predictions, Levinson Durbin Algorithm.

Unit 3

Statistical Decision Theory: Bayes' Criterion, Binary Hypothesis Testing, M-ary Hypothesis Testing, Minimax Criterion, Neyman-Pearson Criterion, Composite Hypothesis Testing.

Parameter Estimation Theory: Maximum Likelihood Estimation, Generalized Likelihood Ratio Test, Some Criteria for Good Estimators, Bayes' Estimation Minimum Mean-Square Error Estimate, Minimum, Mean Absolute Value of Error Estimate Maximum A Posteriori Estimate, Multiple Parameter Estimation Best Linear Unbiased Estimator, Least-Square Estimation Recursive Least-Square Estimator.

Unit 4

Spectral analysis: Estimated autocorrelation function, Periodogram, Averaging the periodogram (Bartlett Method), Welch modification, Parametric method, AR(p) spectral estimation and detection of Harmonic signals.

Information Theory and Source Coding: Introduction, Uncertainty, Information and Entropy, Source coding theorem, Huffman, Shannon-Fano, Arithmetic, Adaptive coding, RLE, LZW Data compaction, LZ-77, LZ-78. Discrete Memory less channels, Mutual information, channel capacity, Channel coding theorem, Differential entropy and mutual information for continuous ensembles.

Unit 5

Application of Information Theory: Group, Ring & Field, Vector, GF

addition,multiplication rules. Introduction to BCH codes, Primitive elements ,Minimal polynomials, Generator polynomials in terms of Minimal polynomials, Some examples of BCH codes,& Decoder, Reed- Solomon codes & Decoder, Implementation of Reed Solomon encoders and decoders.

References:

1. Papoulis and S.U. Pillai, “Probability, Random Variables and Stochastic Processes”,4th Edition, McGraw-Hill, 2002.
2. D.G. Manolakis, V.K. Ingle and S.M. Kogon, “Statistical and Adaptive Signal Processing”, McGraw Hill, 2000.
3. Mourad Barkat , “Signal Detection and Estimation”, Artech House, 2nd Edition, 2005.
4. R G. Gallager, “Information theory and reliable communication”, Wiley, 1st edition, 1968.
5. F. J. MacWilliams and N. J. A. Sloane, “The Theory of Error-Correcting Codes”, NewYork, North-Holland, 1977.
6. Rosen K.H, “Elementary Number Theory”, Addison-Wesley, 6th edition, 2010.

CSPE13: Program Elective-I

3. Wireless Sensor Networks

Instruction Hours/week: 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination : 60

Course Description: This course offers an insight into the concepts of mobile and wireless data communication technologies. The objective of this course is to enable the student to understand the emerging technologies of wireless and mobile communications and simulate them.

Course Objectives:

1. To understand the new trends in mobile/wireless communications networks.
2. To understand multiple radio access techniques.
3. To analyze various routing algorithms used in mobile/wireless networks.
4. To identify the issues in transport and application layers.

Course Outcomes:

At the end of this course, students will be able to

1. Design wireless sensor network system for different applications under consideration.
2. Understand the hardware details of different types of sensors and select right type of sensor for various applications.
3. Understand radio standards and communication protocols to be used for wireless sensor network based systems and application.

4. Use operating systems and programming languages for wireless sensor nodes, performance of wireless sensor networks systems and platforms.
5. Handle special issues related to sensors like energy conservation and security challenges.

Syllabus Contents:

Unit 1

Introduction and overview of sensor network architecture and its applications, sensornetwork comparison with Ad Hoc Networks, Sensor node architecture with hardware and software details.

Hardware: Examples like mica2, micaZ, telosB, cricket, Imote2, tmote, btnode, and Sun

SPOT, Software (Operating Systems): tinyOS, MANTIS, Contiki, and RetOS.

Unit 2

Programming tools: C, nesC. Performance comparison of wireless sensor networks simulation and experimental platforms like open source (ns-2) and commercial (QualNet, Opnet)

Unit 3

Overview of sensor network protocols (details of atleast 2 important protocol per layer):Physical, MAC and routing/ Network layer protocols, node discovery protocols, multi-hop and cluster based protocols, Fundamentals of 802.15.4, Bluetooth, BLE (Bluetooth low energy), UWB.

Unit 4

Data dissemination and processing; differences compared with other database management systems, data storage; query processing.

Unit 5

Specialized features: Energy preservation and efficiency; security challenges; fault-tolerance, Issues related to Localization, connectivity and topology, Sensor deployment mechanisms; coverage issues; sensor Web; sensor Grid, Open issues for future research, and Enabling technologies in wireless sensor network.

References:

1. I.H. Karl and A. Willig, "Protocols and Architectures for Wireless Sensor Networks", John Wiley & Sons, India, 2012.
2. C. S. Raghavendra, K. M. Sivalingam, and T. Znati, Editors, "Wireless Sensor Networks", Springer Verlag, 1st Indian reprint, 2010.
3. F. Zhao and L. Guibas, "Wireless Sensor Networks: An Information Processing Approach", Morgan Kaufmann, 1st Indian reprint, 2013.

4. YingshuLi, MyT. Thai, Weili Wu, "Wireless sensor Network and Applications", Springer
5. Series on signals and communication technology, 2008.

CM14C: Program Elective-I

4. High Performance Networks

Instruction Hours/week: 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination :60

Course Description: This course provides an introduction to voice and data networking technologies, including public and private voice services, Ethernet and Internet data technologies, network security, business applications and net work management. The structure, regulation, and history of the telecom and data network industry will be discussed as well.

Course Objectives:

1. To protocol, algorithms, trade-offs rationale.
2. To routing, transport, DNS resolutions
3. To network extensions and next generation architecture.

Course Outcomes:

At the end of this course, students will be able to

1. Apply knowledge of mathematics, probability, and statistics to model and analyze some networking protocols.
2. Design, implement, and analyze computer networks.
3. Identify, formulate, and solve network engineering problems.
4. Show knowledge of contemporary issues in high performance computer networks. Use techniques, skills, and modern networking tools necessary for engineering practice.

Syllabus Contents:

Unit 1

Types of Networks, Network design issues, Data in support of network design. Network design tools, protocols and architecture. Streaming stored Audio and Video, Best effort service, protocols for real time interactive applications, Beyond best effort, scheduling and policing mechanism, integrated services, and RSVP-differentiated services.

Unit 2

VoIP system architecture, protocol hierarchy, Structure of a voice endpoint, Protocols for the transport of voice media over IP networks. Providing IP quality of service for voice, signaling protocols for VoIP, PSTN gateways, VoIP applications.

Unit 3

VPN-Remote-Access VPN, site-to-site VPN, Tunneling to PPP, Security in VPN. MPLS-operation, Routing, Tunneling and use of FEC, Traffic Engineering, MPLS based VPN, overlay networks-P2P connections.

Unit 4

Traffic Modeling: Little's theorem, Need for modeling, Poisson modeling, Non-poissonmodels, Network performance evaluation.

Unit 5

Network Security and Management: Principles of cryptography, Authentication, integrity, key distribution and certification, Access control and fire walls, attacks and counter measures, security in many layers. Infrastructure for network management, The internet standard management framework –SMI, MIB, SNMP, Security and administration, ASN.1.

References:

1. Kershenbaum A., "Telecommunications Network Design Algorithms", Tata McGraw Hill, 1993.
2. Larry Peterson & Bruce David, "Computer Networks: A System Approach", Morgan Kaufmann, 2003.
3. Douskalis B., "IP Telephony: The Integration of Robust VoIP Services", Pearson Ed. Asia, 2000.
4. Warland J., Varaiya P., "High-Performance Communication Networks", Morgan Kaufmann, 1996.
5. Stallings W., "High-Speed Networks: TCP/IP and ATM Design Principles", Prentice Hall, 1998.
6. Leon Garcia, Widjaja, "Communication networks", TMH 7th reprint 2002.
7. William Stallings, "Network security, essentials", Pearson education Asia publication, 4th Edition, 2011.

CM15C: Program Elective-II

1.Cognitive Radio

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination : 60

Course Description: To understand the fundamental concepts of cognitive radios networks and develop the cognitive radio, as well as techniques for spectrum holes detection that cognitive radio takes advantages in order to exploit it. Technologies to allow an efficient use of TVWS for radio communications based on two spectrum sharing business models.

Course Objectives:

1. To understand the fundamental concepts of cognitive radio networks and develop the cognitive radio, as well as techniques for spectrum holes detection that cognitive radio takes advantages in order to exploit it.
2. Understand technologies to allow an efficient use of TVWS for radio communications based on two spectrum sharing business models/policies

Course Outcomes:

At the end of this course, students will be able to

1. Understand the fundamental concepts of cognitive radio networks.
2. Develop the cognitive radio, as well as techniques for spectrum holes detection that cognitive radio takes advantages in order to exploit it.
3. Understand technologies to allow an efficient use of TVWS for radio communications based on two spectrum sharing business models/policies
4. Understand fundamental issues regarding dynamic spectrum access, the radio-resource management and trading, as well as a number of optimisation techniques for better spectrum exploitation.

Syllabus Contents:**Unit 1**

Introduction to Cognitive Radios: Digital dividend, cognitive radio (CR) architecture, functions of cognitive radio, dynamic spectrum access (DSA), components of cognitive radio, spectrum sensing, spectrum analysis and decision, potential applications of cognitive radio.

Unit 2

Spectrum Sensing: Spectrum sensing, detection of spectrum holes (TVWS), collaborative sensing, geo-location database and spectrum sharing business models (spectrum of commons, real time secondary spectrum market).

Unit 3

Optimization Techniques of Dynamic Spectrum Allocation: Linear programming, convex programming, non-linear programming, integer programming, dynamic programming, stochastic programming.

Unit 4

Dynamic Spectrum Access and Management: Spectrum broker, cognitive radio architectures, centralized dynamic spectrum access, distributed dynamic spectrum access, learning algorithms and protocols.

Unit 5

Spectrum Trading: Introduction to spectrum trading, classification to spectrum trading, radio resource pricing, brief discussion on economics theories in DSA (utility, auction theory), classification of auctions (single auctions, double auctions, concurrent, sequential).

Research Challenges in Cognitive Radio: Network layer and transport layer issues, cross-layer design for cognitive radio networks.

References:

1. Ekram Hossain, Dusit Niyato, Zhu Han, "Dynamic Spectrum Access and Management in Cognitive Radio Networks", Cambridge University Press, 2009.
2. Kwang-Cheng Chen, Ramjee Prasad, "Cognitive radio networks", John Wiley & Sons Ltd., 2009.
3. Bruce Fette, "Cognitive radio technology", Elsevier, 2nd edition, 2009.
4. Huseyin Arslan, "Cognitive Radio, Software Defined Radio, and Adaptive Wireless Systems", Springer, 2007.
5. Francisco Rodrigo Porto Cavalcanti, Soren Andersson, "Optimizing Wireless Communication Systems" Springer, 2009.
6. Linda Doyle, "Essentials of Cognitive Radio", Cambridge University Press, 2009.

CM15C :Program Elective-II

2. Voice and Data Networks

(Common to signal processing)

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination : 60

Course Description: This course provides an introduction to voice and data networking technologies, including design and performance issues. And also focus is on layered & cross layer communication along with Packet and circuit switching communication. Provides Link layer error and flow control mechanisms, also know about queuing models of networks and its applications. Finally have idea about different internet protocols and packet scheduling algorithm.

Course Objectives:

1. Know different design and performance issues of communications networks.
2. Understand the concepts of layered & cross layer communication, along with circuit & packet switching technologies and their deployments in public networks
3. Understand the role of data link layer and various queuing models for

communication.

4. Understand the functions of internetworking devices and importance of different internet protocols.

Course Outcomes:

1. An ability to apply knowledge of networking. network topologies in designing of a network.
2. An ability to model systems using concept of layered/cross layered and TCP/IP architecture.
3. An ability apply the link layer functionalities and queuing models during communication.
4. An ability to use modern engineering techniques for analysis and design of a network with the knowledge of internet concepts and protocols.

Syllabus Contents:

Unit 1

Network Design Issues, Network Performance Issues, Network Terminology, centralized and distributed approaches for networks design, Issues in design of voice and data networks.

Unit 2

Layered and Layer less Communication, Cross layer design of Networks, Voice Networks (wired and wireless) and Switching, Circuit Switching and Packet Switching, Statistical Multiplexing.

Unit 3

Data Networks and their Design, Link layer design- Link adaptation, Link Layer Protocols, Retransmission. Mechanisms (ARQ), Hybrid ARQ (HARQ), Go Back N, Selective Repeat protocols and their analysis.

Unit 4

Queuing Models of Networks , Traffic Models , Little's Theorem, Markov chains, M/M/1 and other Markov systems, Multiple Access Protocols , Aloha System , Carrier Sensing , Examples of Local area networks,

Unit 5

Inter-networking, Bridging, Global Internet, IP protocol and addressing, Sub netting, Classless Inter domain Routing (CIDR) , IP address lookup , Routing in Internet. End to End Protocols, TCP and UDP. Congestion Control, Additive Increase/Multiplicative Decrease, Slow Start, Fast Retransmit/ Fast Recovery, Congestion avoidance, RED TCP Throughput Analysis, Quality of Service in Packet Networks, Network Calculus, Packet Scheduling Algorithms.

References:

1. D. Bertsekas and R. Gallager, "Data Networks", 2nd Edition, Prentice Hall, 1992.
2. L. Peterson and B. S. Davie, "Computer Networks: A Systems Approach", 5th Edition, Morgan Kaufman, 2011.
3. Kumar, D. Manjunath and J. Kuri, "Communication Networking: An analytical approach", 1st Edition, Morgan Kaufman, 2004.
4. Walrand, "Communications Network: A First Course", 2nd Edition, McGraw Hill, 2002.
5. Leonard Kleinrock, "Queueing Systems, Volume I: Theory", 1st Edition, John Wiley and Sons, 1975.
6. Aaron Kershenbaum, "Telecommunication Network Design Algorithms", McGraw Hill, 1993.
7. Vijay Ahuja, "Design and Analysis of Computer Communication Networks", McGraw Hill, 1987

CM15C: Program Elective-II
3. Secured Communication**Instruction Hours/week : 3(L)****Credits : 3****Sessional Marks : 40****Semester-end Examination : 60**

Course Description: The principles are tempered with their practical significance to cope up with the interest to both researchers and system designers. Learning is facilitated by streamlined derivations and assignments.

Course Objectives:

1. To introduce the basic concept encryption techniques
2. To familiarize with the concept of private key and public key cryptosystems.
3. To introduce the concept of Elliptic curves

Course Outcomes: After completing this course, students will be able to:

1. To Understand Cryptography attacks, Integer arithmetic, linear congruence
2. To Understand encryption techniques
3. To Understand Private key and public cryptosystem
4. To Understand Elliptic Curves
5. Discrete logarithm problem on Elliptic Curves

UNIT 1

Introduction: Security Goals, Cryptographic Attacks, Services and Mechanisms, Integer Arithmetic, Modular Arithmetic, Linear Congruence.

UNIT 2

Basic Encryption Techniques: Concept of cryptanalysis, Symmetric key, Block ciphers, Cryptographic algorithms, Features of DES, Stream ciphers, Pseudo random sequence generators, linear complexity, Non-linear combination of LFSRs, Boolean functions.

UNIT 3

Private Key and Public Key Cryptosystems: Asymmetric Key, One way functions, Primality Testing, Factorization problem, Chinese Remainder Theorem, RSA encryption, Diffie Hellmann key exchange, Message authentication and hash functions, Digital signatures, Secret sharing, features of visual cryptography, other applications of cryptography.

UNIT 4

Elliptic Curves: Basic theory, Weirstrass equation, Group law, Point at Infinity, Elliptic curves over finite fields.

UNIT 5

Discrete logarithm problem on Elliptic curve: Elliptic curve cryptography, Diffie Hellmann key exchange over EC, Elgamal encryption over EC, ECDSA.

Text Books:

1. Douglas R. Stinson, Maura B. Paterson "Cryptography, Theory and Practice", 4th Edition, CRC Press, Taylor & Francis Group, 2019.
2. Lawrence C. Washington, "Elliptic Curves", **2nd Edition**, Chapman & Hall/ CRC Press Publishers, April 2008, ISBN: 9781420071474.
3. David S. Dummit, Richard M. Foote, "Abstract Algebra", Third Edition John Wiley & Sons, Inc., 2004.

CM15C: Program Elective-II

4. Electromagnetic Interference And Compatibility

Instruction Hours/Week: 3L

Credits: 3

Sessional Marks

:40

End Semester Examination Marks: 60

Course objective:

1. The course Electromagnetic interference and compatibility gives the in-depth expose of unintentional generation, propagation and reception of electromagnetic energy which may cause unwanted effects such as

electromagnetic interference or even physical damage to the operational or impaired function of equipment.

2. The goal of EMC is to inbuilt the withstanding capability of operation of different equipment in a common electromagnetic environment. This course will equip the learner's in-depth understanding of Sources and victim of EMI. The course gives the insight of the suppression techniques like Shielding, bonding Grounding, Earthing.
3. Learners should be able to explore and apply EMI/EMC measurement technique, evaluation methods and various EMI/EMC standards.
4. A Prerequisite knowledge of Antennas and Microwave Engineering is required for the course.

Syllabus Contents:

UNIT I

BASIC THEORY (10): Introduction to EMI and EMC, Intra and inter system EMI, Elements of Interference, Sources and Victims of EMI, Conducted and Radiated EMI emission and susceptibility, Case Histories, Radiation hazards to humans, Various issues of EMC, EMC Testing categories, EMC Engineering Application. Importance of EMI emission in communications.

UNIT II

EMI from Apparatus and circuits (10): Electromagnetic emissions, noise from relays and switches, non-linearities in circuits, passive inter-modulation, cross-talk in transmission lines, transients in power supply lines, electromagnetic interference(EMI), Overview on Open area test sites and measurements

UNIT III

Radiated and Conducted Interference Measurements (10): Anechoic chamber, TEM cell, GH TEM cell, characterization of conduction Currents/voltages, conducted EM noise on power lines, conducted EMI from Equipment, immunity to conducted EMI detectors and Measurements.

UNIT IV

STANDARDS AND REGULATION (10): Need for Standards, Generic/General Standards for Residential and Industrial environment, Basic Standards, Product Standards, National and International EMI Standardizing Organizations; IEC, ANSI, FCC, CISPR, CENELEC, Electro Magnetic Emission and susceptibility standards and specifications, MIL461E Standards.

UNIT V

EMI TEST METHODS AND INSTRUMENTATION (10): Fundamental

considerations, Basic principles of RE, CE, RS and CS measurements, EMI Shielding effectiveness tests, TEM cell for immunity test, Shielded chamber , Shielded anechoic chamber, EMI test receivers, Spectrum analyzer, EMI test wave simulators, EMI coupling networks, Line impedance stabilization networks, Feed through capacitors, Antennas, Current probes, MIL -STD test methods, Civilian STD test methods.

TEXT BOOK

1. Clayton Paul, "Introduction to Electromagnetic Compatibility", Wiley Interscience, 2006.
2. V Prasad Kodali, "Engineering Electromagnetic Compatibility", IEEE Press, Newyork, 2001.

REFERENCES

1. Henry W. Ott, "Electromagnetic Compatibility Engineering", John Wiley & Sons Inc, Newyork, 2009.
2. Daryl Gerke and William Kimmel, "EDN's Designer's Guide to Electromagnetic Compatibility", Elsevier Science & Technology Books, 2002.
3. W Scott Bennett, "Control and Measurement of Unintentional Electromagnetic Radiation", John Wiley & Sons Inc., (Wiley Interscience Series) 1997.
4. Dr Kenneth L Kaiser, "The Electromagnetic Compatibility Handbook", CRC Press 2005.

COURSE OUTCOMES:

1. Identify the various types and mechanisms of Electromagnetic Interference
2. Identify the various types and mechanisms of Electromagnetic Interference Analyze various sources of EMI and various possibilities to provide EMC.
3. Analyze possible EMI prevention techniques such as grounding, shielding, filtering and use of proper coupling mechanisms to improve compatibility of electronic circuits and systems in a given electromagnetic environment.
4. To measure emission immunity level from different systems to couple with the prescribed EMC standards
5. Understand the different types of EMI/EMC measurement techniques and measuring equipment

CM16L: Advanced Digital Signal Processing Lab (Common to signal Processing)

Instruction Hours/week : 3(P)

Credits : 1.5

Sessional Marks : 40

Semester-end Examination : 60

Course Overview: This practical course enables students to apply skills learned in Advanced Digital Signal Processing algorithms and will help to teach implementation of them. MATLAB is used to apply theoretical concepts and to demonstrate signal processing techniques by means of hands-on application examples. Recognize estimation problems and design, implement and analyze algorithms for solving them. Implement signal processing systems with DSP based development platforms.

Pre-requisites: Signals & Systems, Digital Signal Processing, Digital Signal Processing Lab, MATLAB, and Code Composer Studio.

Course Learning Objectives: The student after studying this course is able to:

1. Understand how to analyze and manipulate digital signals and have the programming knowledge to do so.
2. Understand the trade-offs in the practical design and implement of various structures of FIR and IIR digital filters, for example which can be reduces noise or realizes various applications.
3. Understand theory of prediction and solution of normal equations.
4. Acquaint knowledge of the concepts, algorithms and applications of adaptive signal processing in digital communication systems.
5. Understand the implementation of adaptive filters used in signal processing applications.
6. Understand different methods for estimation of power spectra and its analysis.

Course Outcomes: After completion of this course the student is:

1. Able to design, analyze, and implement digital filters using MATLAB.
2. Able to implement various structures of digital filters.
3. Able to implement up-sampling and down-sampling of a given Sequences.
4. Apply the Lattice filter architecture and implement the Wiener filter, Least Squares, LMS and RLS algorithms, and apply to selected applications.
5. Deduce and apply correlation functions and power spectra for various signal classes, in particular for stochastic signals.

Course Contents:

Note: (i) Minimum of 10 Experiments have to be done invariably.

(ii) All Experiments may be Simulated using MATLAB and to be

verified theoretically.

List of Experiments:

1. Introduction to DSP with MATLAB Programming Environment and Familiarization to Code Composer Studio.
2. Basic Operations on Signals, Generation of Various Signals and finding its FFT.
3. Simple Digital Filters (LPF, HPF, BPF, BSF and Comb Filters).
4. Realization of Structures (Cascade / Parallel/ Lattice Structure's) of a system transfer function.
5. Implement Program to verify Decimation and Interpolation of a given Sequences.
6. Implement Program to Convert CD data into DVD data (sampling rate converters).
7. Estimation of data series using Nth order Forward Predictor and comparing to the Original Signal.
8. Design of linear prediction coding (LPC) filter using Levinson-Durbin Algorithm.
9. Apply the Lattice filter architecture from the Levinson-Durbin algorithm.
10. Computation of Reflection Coefficients using Schur Algorithm.
11. To implement the Wiener filter, and Least Squares algorithms, and apply to selected applications.
12. To implement the LMS and RLS algorithms, and apply to selected applications.
13. Design and verification of a Matched filter.
14. Adaptive Noise Cancellation / Linear Equalizer.
15. Computing Power Spectrum of a Square Signal and chirp signal.
16. Design and Simulation of Notch Filter to remove 50Hz Hum / any unwanted frequency component of given Signal (Speech / ECG).
17. Plot the Periodogram of a Noisy Signal and estimate PSD using Periodogram and Modified Periodogram methods.
18. Estimation of Power Spectrum using Bartlett and Welch methods.
19. Parametric methods (Yule-Walker and Burg) of Power Spectrum Estimation.

For Motivated Learners Extra experiments can be done:

Applications: Adaptive filter and experiments on communication such as generation of a N-tuple PN sequence, generation of a white noise sequence using the PN sequence, restoration of a sinusoidal signal embedded in white noise by Wiener Filtering; speech and multi-media applications.

Texts / References:

1. **John G. Proakis** and **Dimitris G. Manolakis**, Digital Signal Processing:

- Principles, Algorithms and Applications, 5th Edition, **ISBN-13: 9780137348657, Published** by Pearson, 2021.
2. Sanjit Kumar Mitra, Digital Signal Processing: A Computer Based Approach, 4th edition, McGraw Hill Education, 2011.
 3. Ljiljana Milić, Multirate Filtering for Digital Signal Processing: MATLAB Applications, Published by Information Science Reference, Hershey PA 17033, New York, ISBN 978-1-60566-178-0 (Hardcover) - ISBN 978-1-60566-179-7 (eBook), 2009.
 4. R. Chassaing and D. Reay, Digital Signal Processing and Applications with TMS320C6713 and TMS320C6416, Wiley, 2008.

CM17L: Wireless and Mobile Communication Lab

Instruction Hours/week : 3(P)

Credits : 1.5

Sessional Marks : 40

Semester-end Examination : 60

Course Objectives:

1. To enable the students to understand and investigate Wireless channels, Fading environment and analyze their behaviour.
2. To study various radio propagation models.
3. To describe different algorithms of adaptive filtering.
4. To facilitate students with hands on 4G LTE network to establish communication between user equipment.

List of Experiments

1. Radio propagation models: Okumura and Hata models.
2. Calculation of frequency reuse factor and cluster size.
3. Improving channel capacity of wireless network using cell sectoring.
4. Selection diversity for wireless communication.
5. Power spectral density of GMSK.
6. RLS and LMS algorithms for adaptive filtering.
7. Configuration of processor and IP address for 4G LTE operation.
8. Configuration of eNodeB Radio parameters and Bandwidth.
9. Configuration of EPC and user database.
10. Configuration of IMS and users, complete LTE run and UE application setup.
11. Virtual lab experiments on understanding of pathloss, beam patterns, calculation of boundary coverage probability and SINR.
12. Virtual lab experiments on handoff, frequency reuse, flat and frequency selective fading.

Course Outcomes:

Upon completion of the course, students will be able to

1. Understand Cellular concepts and radio propagation models.
2. Perform 4G LTE wireless network through experimentation
3. Analyze configuration parameters of Software Defined Radio in real time environment.
4. Demonstrate various concepts of wireless and mobile communication like handoff, cell sectoring, frequency reuse, etc., using virtual labs.

Resources:

1. Software Tools: Matlab
2. Amitec 4G LTE Network Setup
3. Virtual Labs: Fading Channels and Mobile Communications,
<http://vlabs.iitkgp.ernet.in/fcmc>

CM18C: Research Methodology and IPR**Instruction Hours/week : 3L****Credits:3****Sessional Marks : 40****Semester-end Examination : 60**

Description of the Course: This subject gives how to proceed systematically for research, present research findings. This course consists of basics of research methods, paper writing, patenting methods and requirements.

Course Educational Objectives (CEOs):

1. To understand the importance of research objectives and procedures.
2. To know the procedures of data collection and report writing of research.
3. To have the knowledge of filing and obtaining a patent on research findings.

UNIT I

Meaning, Objective and Motivation in Research: Types of Research, Research Approaches, Research Process, Validity and Reliability in Research, Research Design: Features of Good Design, Types of Research Design, Basic Principles of Experimental Design

UNIT II

Sampling Design: Steps in Sampling Design, Characteristics of a Good Sample Design, Random Samples and Random Sampling Design Measurement and Scaling Techniques: Errors in Measurement, , Scaling and Scale Construction techniques, Forecasting Techniques, Time Series

Analysis, Interpolation and Extrapolation

UNIT III

Methods of Data Collection: Primary Data, Questionnaire and Interviews, Collection of Secondary Data, Cases and Schedules. Professional Attitude and Goals, Concept of Excellence, Ethics in Science and Engineering, Correlation and Regression Analysis, Method of Least Squares, Regression Vs. Correlation, Correlation Vs. Determination.

UNIT IV

Interpretation of Data and Report Writing, Layout of a Research Paper, Techniques of Interpretation. Making Scientific Presentation at Conferences and Popular Lectures to Semi Technical Audience, Participating in Public Debates on Scientific Issues

UNIT V

Nature of Intellectual property rights, Patents, designs, trademarks and copyrights, History of patenting, process of patenting, patent development, international cooperation on IPR, procedure of granting patent, patent rights, licensing and transferring technology, Geographical Indications ,IPR in biological and systems and software.

Text/Reference Books:

1. Research Methodology: Methods And Techniques - C. R. Kothari, 2nd Edition, New Age International Publishers.
2. Statistical Methods - S P. Gupta. S. Chand & Sons, New Delhi.
3. Intellectual Property-the law of trademarks, copyrights, Patents, and trade Secrets- Deborah E. Bouchoux, Esq. Georgetown University, Fourth edition

Course Outcomes (COs):

1. Able to know research importance and requirements and procedure
2. Able to apply sampling techniques for analysis and forecasting.
3. Able to use methods for data collection and analyze the data using different mathematical techniques.
4. Able to write report on research done and present the research findings in systematic manner.
5. Able to have the knowledge to file and obtain patent on research finding.

Semester II

CM21C: Antennas and Radiating Systems

Instruction Hours/week : (L-T-P-C): 3-1-0-4

Credits : 4

Sessional Marks : 40

Semester-end Examination :60

Course Description: To understand the fundamental principles of antenna theory and to apply them to analysis and design. Because there are so many methods of analysis and design and different antenna structures, applications are made to some of the most basic and practical configurations such as linear dipoles, loops, arrays, aperture antennas, horn antennas, microstrip antennas and reflector antennas.

Course Objectives:

1. To fundamental antenna parameters and numerical methods to analyze and differentiate the antennas.
2. To concept of radiation mechanism of various antennas.
3. To mechanism and models for radio-wave propagation.

Course Outcomes:

At the end of this course, students will be able to

1. Compute the far field distance, radiation pattern and gain of an antenna for given current distribution.
2. Estimate the input impedance, efficiency and ease of match for antennas.
3. Compute the array factor for an array of identical antennas.
4. Design antennas and antenna arrays for various desired radiation pattern characteristics.

Syllabus Contents:

Unit 1

Types of Antennas: Wire antennas, Aperture antennas, Micro strip antennas, Arrayantennas Reflector antennas, Lens antennas, Radiation Mechanism, Current distribution on thin wire antenna.

Fundamental Parameters of Antennas: Radiation Pattern, Radiation Power Density, Radiation Intensity, Directivity, Gain, Antenna efficiency, Beam efficiency, Bandwidth, Polarization, Input Impedance, radiation efficiency, Antenna Vector effective length, Friis Transmission equation, Antenna Temperature.

Unit 2

Linear Wire Antennas: Infinitesimal dipole, Small dipole, Region separation, Finite lengthdipole, half wave dipole, Ground effects.

Loop Antennas: Small Circular loop, Circular Loop of constant current, Circular loop with non uniform current.

Unit 3

Linear Arrays: Two element array, N Element array: Uniform Amplitude and spacing, Broadside and End fire array, Super directivity, Planar array, Design consideration.

Unit 4

Aperture Antennas: Huygen's Field Equivalence principle, radiation equations, Rectangular Aperture, Circular Aperture.

Horn Antennas: E-Plane, H-plane Sectoral horns, Pyramidal and Conical horns.

Unit 5

Micro strip Antennas: Basic Characteristics, Feeding mechanisms, Method of analysis, Rectangular Patch, Circular Patch, Reflector Antennas: Plane reflector, parabolic reflector, Cassegrain reflectors, Introduction to MIMO.

References:

1. Constantine A. Balanis, "Antenna Theory Analysis and Design", John Wiley & Sons, 4th edition, 2016.
2. John D Kraus, Ronald J Marhefka, Ahmad S Khan, "Antennas for All Applications", Tata McGraw-Hill, 2002.
3. R.C. Johnson and H. Jasik, "Antenna Engineering hand book", Mc-Graw Hill, 1984.
4. I.J. Bhal and P. Bhartia, "Micro-strip antennas", Artech house, 1980.

CM22C: Advanced Communication Networks

Instruction Hours/week : (L-T-P-C): 3-1-0-4

Credits :4

Sessional Marks : 40

Semester-end Examination :60

Description of the Course: Basic techniques for modeling and analyzing communication networks. Fairness and utility functions, routing, congestion control, pricing, queuing models, loss networks, multi-class queues and scheduling.

Course Objectives:

1. To fairness and network utility maximization
2. To optimization based routing and congestion control
3. To basic queuing models and their application to switching and scheduling in networks.

Course Outcomes:

At the end of this course, students will be able to

1. Understand advanced concepts in Communication Networking.
2. Design and develop protocols for Communication Networks.
3. Understand the mechanisms in Quality of Service in networking.
4. Optimise the Network Design

Syllabus Contents:

Unit 1

Overview of Internet-Concepts, challenges and history. Overview of -ATM. TCP/IP Congestion and Flow Control in Internet-Throughput analysis of TCP congestion control. TCP for high bandwidth delay networks. Fairness issues in TCP.

Unit 2

Real Time Communications over Internet. Adaptive applications. Latency and throughput issues. Integrated Services Model (IntServ). Resource reservation in Internet. RSVP.

Characterization of Traffic by Linearly Bounded Arrival Processes (LBAP). Leaky bucket algorithm and its properties.

Unit 3

Packet Scheduling Algorithms-requirements and choices. Scheduling guaranteed service connections. GPS, WFQ and Rate proportional algorithms. High speed scheduler design. Theory of Latency Rate servers and delay bounds in packet switched networks for LBAP traffic.

Active Queue Management - RED, WRED and Virtual clock. Control theoretic analysis of active queue management.

Unit 4

IP address lookup-challenges. Packet classification algorithms and Flow Identification-Grid of Tries, Cross producting and controlled prefix expansion algorithms. Admission control in Internet. Concept of Effective bandwidth. Measurement based admission control. Differentiated Services in Internet (DiffServ). DiffServ architecture and framework.

Unit 5

IPv4, IPv6, IP tunnelling, IP switching and MPLS, Overview of IP over ATM and its evolution to IP switching. MPLS architecture and framework. MPLS Protocols. Traffic engineering issues in MPLS.

References:

1. Jean Wairand and Pravin Varaiya, "High Performance Communications Networks", 2nd edition, 2000.
2. Jean Le Boudec and Patrick Thiran, "Network Calculus A Theory of Deterministic Queueing Systems for the Internet", Springer Verlag,

- 2001.
3. Zhang Wang, "Internet QoS", Morgan Kaufman, 2001.
 4. Anurag Kumar, D. Manjunath and Joy Kuri, "Communication Networking: An Analytical Approach" , Morgan Kaufman Publishers, 2004.
 5. George Kesidis, "ATM Network Performance", Kluwer Academic, Research Papers, 2005.

CM23C: Program Elective-III

1. Satellite Communication

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination :60

Description of the Course: Antennas and propagation effects play a crucial role in RF systems. In practice, the design of a working system such as mobile phone networks, WiFi, RFID, Satellite communication and GPS requires a good understanding of these components. This course teaches the fundamentals of antenna and propagation and shows the application in practical examples. The course covers the theory of radiation, fundamental antenna parameters and concepts, wire antennas such as dipoles and loop antennas, antenna arrays, aperture antennas, microstrip antennas, numerical analysis, communication & radar systems and propagation effects.

Course Objectives:

1. To fairness and network utility maximization
2. To optimization based routing and congestion control
3. To basic queuing models and their application to switching and scheduling in networks.

Course Outcomes:

At the end of this course, students will be able to

1. Visualize the architecture of satellite systems as a means of high speed, high range communication system.
2. State various aspects related to satellite systems such as orbital equations, sub-systems in a satellite, link budget, modulation and multiple access schemes.
3. Solve numerical problems related to orbital motion and design of link budget for the given parameters and conditions.

Syllabus Contents:

Unit 1:

Architecture of Satellite Communication System: Principles and architecture

of satellite Communication, Brief history of Satellite systems, advantages, disadvantages, applications, and frequency bands used for satellite communication and their advantages/drawbacks.

Unit 2:

Orbital Analysis: Orbital equations, Kepler's laws of planetary motion, Apogee and Perigee for an elliptical orbit, evaluation of velocity, orbital period, angular velocity etc of a satellite, concepts of Solar day and Sidereal day.

Unit 3:

Satellite sub-systems: Architecture and Roles of various sub-systems of a satellite system such as Telemetry, tracking, command and monitoring (TTC & M), Attitude and orbit control system (AOCS), Communication sub-system, power sub-systems, antenna sub-system.

Unit 4:

Typical Phenomena in Satellite Communication: Solar Eclipse on satellite, its effects, remedies for Eclipse, Sun Transit Outage phenomena, its effects and remedies, Doppler frequency shift phenomena and expression for Doppler shift.

Unit 5:

Satellite link budget: Flux density and received signal power equations, Calculation of System noise temperature for satellite receiver, noise power calculation, Drafting of satellite link budget and C/N ratio calculations in clear air and rainy conditions, Case study of Personal Communication system (satellite telephony) using LEO. Modulation and Multiple Access Schemes used in satellite communication. Typical case studies of VSAT, DBS-TV satellites and few recent communication satellites launched by NASA/ ISRO. GPS.

References:

1. Timothy Pratt and Others, "Satellite Communications", Wiley India, 2nd edition, 2010.
2. S. K. Raman, "Fundamentals of Satellite Communication", Pearson Education India, 2011. Tri T. Ha, "Digital Satellite Communications", Tata McGraw Hill, 2009.
3. Dennis Roddy, "Satellite Communication", McGraw Hill, 4th Edition, 2008.

CM23C: Program Elective-III
2. IOT and Applications
(Common to Signal processing)

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination :60

Description of the Course: Internet of Things (IoT) cuts across different application domain verticals ranging from civilian to defense sectors. These domains include agriculture, space, healthcare, manufacturing, construction, water, and mining, which are presently transitioning their legacy infrastructure to support a lot. Today it is possible to envision pervasive connectivity, storage, and computation, which, in turn, gives rise to building different lot solutions. IoT-based applications such as innovative shopping systems, infrastructure management in both urban and rural areas, remote health monitoring and emergency notification systems, and transportation systems, are gradually relying on IoT-based systems. Therefore, it is very important to learn the fundamentals of this emerging technology.

Course Objectives:

1. To apprise students with basic knowledge of IoT that paves a platform to understand physical and logical design of IOT
2. To teach a student how to analyse requirements of various communication models and protocols for cost-effective design of IoT applications on different IoT platforms.
3. To introduce the technologies behind Internet of Things (IoT).
4. To explain the students how to code for an IoT application using Arduino/Raspberry Pi open platform.

Syllabus Contents:

UNIT I

THE INTERNET OF THINGS TODAY, TIME FOR CONVERGENCE

M2M to IoT – A Basic Perspective– Introduction, Some Definitions, M2M Value Chains, IoT Value Chains, An emerging industrial structure for IoT, The international driven global value chain and global information monopolies. M2M to IoT-An Architectural Overview– Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations.

UNIT II

COMPONENTS IN INTERNET OF THINGS

Functional Blocks of an IoT Ecosystem – Sensors, Actuators, and Smart Objects – Control Units - Communication modules (Bluetooth, Zigbee,Wi-Fi, GPS, GSM Modules)

UNIT III

PROTOCOLS AND TECHNOLOGIES BEHIND IOT

IOT Protocols - IPv6, 6LoWPAN, MQTT, CoAP - RFID, Wireless Sensor Networks, BigData Analytics, Cloud Computing, Embedded Systems.

UNIT IV

OPEN PLATFORMS AND PROGRAMMING

IOT deployment for Raspberry Pi /Arduino platform-Architecture – Programming – Interfacing – Accessing GPIO Pins – Sending and Receiving Signals Using GPIO Pins – Connecting to the Cloud.

UNIT V

IOT APPLICATIONS

Business models for the internet of things, Smart city, Smart mobility and transport, Industrial IoT, Smart health, Environment monitoring and surveillance – Home Automation – Smart Agriculture

COURSE OUTCOMES:

Upon completion of the course, students will be able to

1. Explain the concept of IoT.
2. Understand the communication models and various protocols for IoT.
3. Design portable IoT using Arduino/Raspberry Pi/open platform.
4. Apply data analytics and use cloud offerings related to IoT.
5. Analyze applications of IoT in real time scenario.

TEXT BOOKS:

1. Dr. Ovidiu Vermesan, Dr. Peter Friess, "Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems", River Publishers, Aalborg, 2013.
2. Robert Barton, Patrick Grossetete, David Hanes, Jerome Henry, Gonzalo Salgueiro, "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things", CISCO Press, 2017.
3. Samuel Greengard, The Internet of Things, The MIT Press, 2015.

REFERENCES:

1. Perry Lea, "Internet of things for architects", Packt, 2018
2. Olivier Hersent, David Boswarthick, Omar Elloumi, "The Internet of Things – Key applications and Protocols", Wiley, 2012
3. IOT (Internet of Things) Programming: A Simple and Fast Way of

Learning, IOT Kindle Edition.

4. Dieter Uckelmann, Mark Harrison, Michahelles, Florian (Eds), "Architecting the Internet of Things", Springer, 2011.
5. ArshdeepBahga, Vijay Madiseti, "Internet of Things – A hands-on approach", Universities Press, 2015.
6. https://www.arduino.cc/https://www.ibm.com/smarterplanet/us/en/?ca=v_smarterplanet.

CM23C: Program Elective-III

3. RF and Microwave Circuit Design

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination :60

Course Objectives:

1. To understand the behaviour of RF passive components model active components and Perform transmission line analysis.
2. To know the use of Smith Chart for high frequency circuit design.
3. To Justify the choice/selection of components from the design aspects and contribute in the areas of RF circuit design

Course Outcomes: At the end of this course, students will be able to

1. Understand the behaviour of RF passive components and model active components.
2. Perform transmission line analysis.
3. Demonstrate use of Smith Chart for high frequency circuit design.
4. Justify the choice/selection of components from the design aspects.
5. Contribute in the areas of RF circuit design.

Syllabus Contents:

Unit 1

Transmission Line Theory: Lumped element circuit model for transmission line, field analysis, Smith chart, quarter wave transformer, generator and load mismatch, impedance matching and tuning.

Microwave Network Analysis: Impedance and equivalent voltage and current, Impedance and admittance matrix, The scattering matrix, transmission matrix, Signal flow graph.

Unit 2

Microwave Components: Microwave resonators, Microwave filters, power dividers and directional couplers, Ferromagnetic devices and components.

Unit 3

Nonlinearity And Time Variance Inter-symbol interference, random process & noise, definition of sensitivity and dynamic range, conversion gain and distortion.

Unit 4

Microwave Semiconductor Devices and Modeling: PIN diode, Tunnel diodes, Varactor diode, Schottky diode, IMPATT and TRAPATT devices, transferred electron devices, Microwave BJTs, GaAs FETs, low noise and power GaAs FETs, MESFET, MOSFET, HEMT.

Unit 5

Amplifiers Design: Power gain equations, stability, impedance matching, constant gain and noise figure circles, small signal, low noise, high power and broadband amplifier, oscillators, Mixers design.

References:

1. Matthew M. Radmanesh, "Advanced RF & Microwave Circuit Design: The Ultimate Guide to Superior Design", AuthorHouse, 2009.
2. D.M.Pozar, "Microwave engineering", Wiley, 4th edition, 2011.
3. R.Ludwig and P.Bretchko, "R. F. Circuit Design", Pearson Education Inc, 2009.
4. G.D. Vendelin, A.M. Pavo, U. L. Rohde, "Microwave Circuit Design Using Linear And Non Linear Techniques", John Wiley 1990.
5. S.Y. Liao, "Microwave circuit Analysis and Amplifier Design", Prentice Hall 1987. Radmanesh, "RF and Microwave Electronics Illustrated", Pearson Education, 2004. W.H. Freeman & Co, 1978

CM23C: Program Elective-III

4. MIMO Systems

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination :60

Course Objectives:

1. To understand channel modelling and propagation, MIMO Capacity, space-time coding MIMO receivers, MIMO for multi-carrier
2. systems (e.g. MIMO-OFDM), multi-user communications, multi-user MIMO.
3. To gain knowledge understand cooperative and coordinated multi-cell MIMO, introduction to MIMO in 4G (LTE, LTE-Advanced, WiMAX).
4. To perform Mathematical modelling and analysis of MIMO systems. To build a small low-cost embedded system using Arduino / Raspberry Pi or

equivalent boards.

Course Outcomes:

At the end of this course, students will be able to

1. Understand channel modelling and propagation, MIMO Capacity, space-time coding.
2. MIMO receivers, MIMO for multi-carrier systems (e.g. MIMO-OFDM), multi-user communications, multi-user MIMO.
3. Understand cooperative and coordinated multi-cell MIMO, introduction to MIMO in 4G (LTE, LTE-Advanced, WiMAX).
4. Perform Mathematical modelling and analysis of MIMO systems.

Syllabus Contents:

Unit 1:

Introduction to Multi-antenna Systems, Motivation, Types of multi-antenna systems, MIMO vs. multi-antenna systems. Diversity, Exploiting multipath diversity, Transmit diversity, Space-time codes, The Alamouti scheme, Delay diversity, Cyclic delay diversity, Space-frequency codes, Receive diversity, The rake receiver, Combining techniques, Spatial Multiplexing, Spectral efficiency and capacity, Transmitting independent streams in parallel, Mathematical notation

Unit 2:

The generic MIMO problem, Singular Value Decomposition, Eigenvalues and eigenvectors, Equalising MIMO systems, Disadvantages of equalising MIMO systems, Pre-distortion in MIMO systems, Disadvantages of pre-distortion in MIMO systems, Pre-coding and combining in MIMO systems, Advantages of pre-coding and combining, Disadvantages of pre-coding and combining, Channel state information.

Unit 3

Codebooks for MIMO, Beamforming, Beamforming principles, Increased spectrum efficiency, Interference cancellation, Switched beamformer, Adaptive beamformer, Narrowband beamformer, Wideband beamformer

Unit 4:

Case study: MIMO in LTE, Codewords to layers mapping, Pre-coding for spatial multiplexing, Pre-coding for transmit diversity, Beamforming in LTE, Cyclic delay diversity based pre-coding, Pre-coding codebooks, Propagation Channels, Time & frequency channel dispersion, AWGN and multipath propagation channels, Delay spread values and time variations, Fast and slow fading environments, Complex baseband multipath channels, Narrowband and wideband channels, MIMO channel models

Unit 5:

Channel Estimation, Channel estimation techniques, Estimation and tracking, Trainingbased channel estimation, Blind channel estimation, Channel estimation architectures, Iterative channel estimation, MMSE channel estimation, Correlative channel sounding, Channel estimation in single carrier systems, Channel estimation for CDMA, Channel estimation for OFDM.

References:

1. Claude Oestges, Bruno Clerckx, "MIMO Wireless Communications : From Real-world Propagation to Space-time Code Design", Academic Press, 1st edition, 2010.
2. Mohinder Janakiraman, "Space - Time Codes and MIMO Systems", Artech House Publishers, 2004.

CM24C: Program Elective-IV
**1. Random Processes and Queuing Models
(Common to Signal Processing)**
Instruction Hours/week : 3(L)
Credits : 3
Sessional Marks : 40
Semester-end Examination :60
Prerequisites: Probability Theory and Stochastic Processes

Preamble: This course is designed to provide necessary basic concepts in random processes which are widely applied in random signals, linear systems in communication engineering and IT fields. The objective of this course is to familiarize the understanding of stochastic processes and queuing models. The syllabus also covers the concepts of Markovian and advanced queueing models which are essential to design and analyze communication networks.

Course Learning Objectives: The objectives of this course is to enable the student to

1. Characterize stochastic processes with an emphasis on stationary random processes.
2. Use the properties of random processes with LTI systems in real world situations.
3. Understand Markov chains and their transient behavior.
4. Differentiate between different models of queuing theory and their performance measures.
5. Understand network of queues with Poisson external arrivals, exponential service requirements and independent routing.

Course Outcomes:

At the end of the course, the students will be able to

1. Analyze the various classifications of Random Processes and characterize phenomena which evolve with respect to time in a probabilistic manner.
2. Apply the ideas of Random Processes to the LTI Systems for Spectral analysis.
3. Understand Markov Chains and regenerative processes used in modeling a wide variety of systems.
4. Understand the basic characteristic features of a queuing system and acquire skills in analyzing queuing models.
5. Analyze the performance of the queuing networks.

UNIT I

REVIEW OF RANDOM PROCESSES: Classification of General Random Processes, Binomial Processes, Poisson Processes, Ergodic Process, Gaussian Random Processes, Stationary and Wide Sense Stationary Random Processes, Random walks and gambler's ruin, Processes with independent increments and martingales, Brownian motion, Counting processes and the Poisson process, Stationarity, Joint properties of random processes.

UNIT II

RANDOM PROCESSES IN LINEAR SYSTEMS AND SPECTRAL ANALYSIS: Basic definitions, Spectral Density Function, transfer functions and power spectral densities, Discrete-time processes in linear systems, Low Pass and Band Pass Processes, Baseband random processes, Narrowband random processes.

UNIT III

MARKOV PROCESSES: Markov Chains, Probability Distribution of a Markov Chain, Transition Probability Matrices of a Markov Chain, Classification of States and Chains, Chapman-Kolmogorov Theorem, Stationary Distribution for a Markov Chains, Classification of States of a Markov Chain, Birth and Death Processes, Renewal Process, The Transition Probability Function, Limiting Probabilities, Exponential Distribution & Poisson Process.

UNIT IV

QUEUEING THEORY: Basic Characteristics of Queueing Models, Introduction to Markovian queueing models, Steady state distribution, Little's Theorem, Cost equations, steady state probabilities, Queueing Models: **(M/M/1) : (∞ /FIFO)** Single server with infinite system capacity, **(M/M/1) : (k/FIFO)**

Single Server with Finite Capacity, **(M/M/s) : (∞ /FIFO)** Multiple Server with Infinite Capacity, **(M/M/s) : (k/FIFO)** Multiple Server with Finite Capacity, models balance equations, Erlang's B and C formulae, M/G/1 Queuing system characteristics, Queues with finite waiting rooms – Queues with impatient customers: Balking and reneging.

Unit V

QUEUEING NETWORKS: Network of queues basic concepts, Tandem Queues, Channels in Series or Jackson Networks, Queues in series with multiple channels at each phase, Closed Jackson Networks, Approximating Closed Networks, Open Networks with General Customer Routes, Symmetric Queues.

Text Books:

1. Sheldon M. Ross, Introduction to Probability Models, 10th Edition, Elsevier, 2010.
2. T Veera Rajan , Probability, Statistics and Random Process, 3rd Edition, Tata Mc Graw Hill, 2008.
3. R. D. Yates and D. J. Goodman, "Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers", 3rd Edition International Student Version, Wiley, 2014.

Reference Books:

1. J.F. Shortle, J.M. Thompson, D. Gross and C.M. Harris, Fundamentals of Queueing Theory, 5th Edition, Wiley, 2018.
2. J. Medhi, Stochastic Models in Queueing Theory, 2nd Edition, Academic Press, 2003.
3. U.N. Bhat, An Introduction to Queueing Theory, Springer, 2015.

CM24C: Program Elective-IV

2. Pattern Recognition and Machine Learning (Common to Signal Processing)

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination :60

Course Objectives:

1. To study the parametric and linear models for classification and design neural network and SVM for classification
2. To develop machine independent and unsupervised learning techniques.

Course Outcomes:

At the end of this course, students will be able to

1. Study the parametric and linear models for classification
2. Design neural network and SVM for classification

3. Develop machine independent and unsupervised learning techniques.

Syllabus Contents:

Unit 1

Introduction to Pattern Recognition: Problems, applications, design cycle, learning and adaptation, examples, Probability Distributions, Parametric Learning - Maximum likelihood and Bayesian Decision Theory- Bayes rule, discriminant functions, loss functions and Bayesian error analysis

Unit 2

Linear models: Linear Models for Regression, linear regression, logistic regression Linear Models for Classification

Unit 3

Neural Network: perceptron, multi-layer perceptron, back propagation algorithm, error surfaces, practical techniques for improving back propagation, additional networks and training methods, Adaboost, Deep Learning

Unit 4

Linear discriminant functions - decision surfaces, two-category, multi-category, minimum-squared error procedures, the Ho-Kashyap procedures, linear programming algorithms, Support vector machine

Unit 5

Algorithm independent machine learning – lack of inherent superiority of any classifier, bias and variance, re-sampling for classifier design, combining classifiers

Unsupervised learning and clustering – k-means clustering, fuzzy k-means clustering, hierarchical clustering

References:

1. Richard O. Duda, Peter E. Hart, David G. Stork, "Pattern Classification", 2nd Edition John Wiley & Sons, 2001.
2. Trevor Hastie, Robert Tibshirani, Jerome H. Friedman, "The Elements of Statistical Learning", 2nd Edition, Springer, 2009.
3. C. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.

CM24C: Program Elective-IV
3. Programmable Networks - SDN, NFV

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination :60

Course Objectives:

1. To understand advanced concepts in Programmable Networks, Software Defined Networking, an emerging Internet architectural framework.
2. To know the main concepts, architectures, algorithms, protocols and applications in SDN and NFV.

Course Outcomes:

At the end of this course, students will be able to

1. Understand advanced concepts in Programmable Networks.
2. Understand Software Defined Networking, an emerging Internet architectural framework.
3. Implement the main concepts, architectures, algorithms, protocols and applications in SDN and NFV.

Syllabus Contents:

Unit 1

Introduction to Programmable Networks, History and Evolution of Software Defined Networking (SDN), Fundamental Characteristics of SDN, Separation of Control Plane and Data Plane, Active Networking.

Unit 2

Control and Data Plane Separation: Concepts, Advantages and Disadvantages, the basics of Open Flow protocol.

Network Virtualization: Concepts, Applications, Existing Network Virtualization Framework, Mininet A simulation environment for SDN.

Unit 3

Control Plane: Overview, Existing SDN Controllers including Floodlight and Open Day light projects. Customization of Control Plane: Switching and Firewall Implementation using SDN Concepts. Data Plane: Software-based and Hardware-based; Programmable Network Hardware.

Unit 4

Programming SDNs: Northbound Application Programming Interface, Current Languages and Tools, Composition of SDNs. Network Functions Virtualization (NFV) and Software Defined Networks: Concepts, Implementation and Applications.

Unit 5

Data Center Networks: Packet, Optical and Wireless Architectures, Network Topologies. Use Cases of SDNs: Data Centers, Internet Exchange Points, Backbone Networks, Home Networks, Traffic Engineering.

References:

1. Thomas D. Nadeau, Ken Gray, "SDN: Software Defined Networks, An Authoritative Review of Network Programmability Technologies", O'Reilly Media, August 2013.
2. Paul Goransson, Chuck Black, Timothy Culver. "Software Defined Networks: A Comprehensive Approach", Morgan Kaufmann Publishers, 2016.
3. Fei Hu, "Network Innovation through OpenFlow and SDN: Principles and Design", CRC Press, 2014.
4. Vivek Tiwari, "SDN and OpenFlow for Beginners", Amazon Digital Services, Inc., ASIN: ,2013.
5. Nick Feamster, Jennifer Rexford and Ellen Zegura, "The Road to SDN: An Intellectual History of Programmable Networks" ACM CCR April 2014.
6. Open Networking Foundation (ONF) Documents, <https://www.opennetworking.org>,2015.<http://www.openflow.org>,

CM24C: Program Elective-IV

4. Remote Sensing

(Common to Signal Processing)

Instruction Hours/week : 3(L)

Credits : 3

Sessional Marks : 40

Semester-end Examination :60

Course Objectives:

1. To understand basic concepts, principles and applications of remote sensing, particularly the geometric and radiometric principles
2. To know the applications of principles to a variety of topics in remote sensing, particularly related to data collection, radiation, resolution, and sampling.

Course Outcomes:

At the end of this course, students shall be able to

1. Understand basic concepts, principles and applications of remote sensing, particularly the geometric and radiometric principles
2. Provide examples of applications of principles to a variety of topics in remote sensing, particularly related to data collection, radiation, resolution, and sampling.

Syllabus Contents:

Unit 1

Physics Of Remote Sensing: Electro Magnetic Spectrum, Physics of Remote Sensing-Effects of Atmosphere-Scattering-Different types-Absorption-Atmospheric window-Energy interaction with surface features –Spectral reflectance of vegetation, soil and water atmospheric influence on spectral response patterns-multi concept in Remote sensing.

Data Acquisition: Types of Platforms–different types of aircrafts-Manned and Unmanned space crafts–sun synchronous and geo synchronous satellites – Types and characteristics of different platforms –LANDSAT, SPOT, IRS, INSAT, IKONOS, QUICKBIRDetc.

Unit 2

Photographic products, B/W,color, color IR film and their characteristics – resolving power of lens and film - Optomechanical electro optical sensors – across track and along track scanners-multispectral scanners and thermal scanners–geometric characteristics of scanner imagery - calibration of thermal scanners.

Unit 3

Scattering System: Microwave scatterometry, types of RADAR –SLAR – resolution –range and azimuth –real aperture and synthetic aperture RADAR. Characteristics of Microwave images topographic effect-different types of Remote Sensing platforms –airborne and space borne sensors -ERS, JERS, RADARSAT, RISAT -Scatterometer, Altimeter-LiDAR remote sensing, principles, applications.

Unit 4

Thermal And Hyper Spectral Remote Sensing: Sensors characteristics-principle of spectroscopy-imaging spectroscopy-field conditions, compound spectral curve, Spectral library, radiative models, processing procedures, derivative spectrometry, thermal remote sensing –thermal sensors, principles, thermal data processing, applications.

Unit 5

Data Analysis: Resolution–Spatial, Spectral, Radiometric and temporal resolution-signal to noise ratio-data products and their characteristics-visual and digital interpretation-Basic principles of data processing – Radiometric correction–Image enhancement–Image classification–Principles of LiDAR, Aerial Laser Terrain Mapping.

References:

1. Lillesand, T.M. and Kiefer, R.W., "Remote Sensing and Image Interpretation", 6th Edition, John Wiley & Sons, 2000.
2. John R. Jensen, "Introductory Digital Image Processing: A Remote Sensing Perspective", 2nd Edition, Prentice Hall, 1995.
3. Richards, John A., Jia, Xiuping, "Remote Sensing Digital Image Analysis", 5th Edition, Springer-Verlag Berlin Heidelberg, 2013.
4. Paul Curran P.J. Principles of Remote Sensing, 1st Edition, Longman Publishing Group, 1984.
5. Charles Elachi, Jakob J. van Zyl, "Introduction to The Physics and Techniques of Remote Sensing", 2nd Edition, Wiley Series, 2006.
6. Sabins, F.F. Jr, "Remote Sensing Principles and Image Interpretation", 3rd Edition, W.H. Freeman & Co, 1978

CM25L: Antennas and Radiating Systems Lab**Instruction Hours/week : 3(P)****Credits : 1.5****Sessional Marks : 40****Semester-end Examination : 60****Course Objectives:**

1. To determine specifications, design, construct and test antenna.
2. To explore and use tools for designing, analyzing and testing antennas. These tools include Antenna design and analysis software, network analyzers, spectrum analyzers, and antenna pattern measurement techniques.

Course Outcomes:

At the end of this course, students will be able to

1. Determine specifications, design, construct and test antenna.
2. Explore and use tools for designing, analyzing and testing antennas. These tools include Antenna design and analysis software, network analyzers, spectrum analyzers, and antenna pattern measurement techniques.

List of Assignments:

1. Simulation of half wave dipole antenna.
2. Simulation of change of the radius and length of dipole wire on frequency of resonance of antenna.
3. Simulation of quarter wave, full wave antenna and comparison of their parameters.
4. Simulation of monopole antenna with and without ground plane.
5. Study the effect of the height of the monopole antenna on the radiation characteristics of the antenna.

6. Simulation of a half wave dipole antenna array.
7. Study the effect of change in distance between elements of array on radiation pattern of dipole array.
8. Study the effect of the variation of phase difference 'beta' between the elements of the array on the radiation pattern of the dipole array.
9. Case study.

CM26L: Advanced Communication Networks Lab

Instruction Hours/week : 3(P)

Credits : 1.5

Sessional Marks : 40

Semester-end Examination :60

Course Objectives:

1. To identify the different types of network devices and their functions within a network.
2. To understand the skills of sub-netting and routing mechanisms, basic protocols of computer networks, and how they can be used to assist in network design and implementation.

Course Outcomes:

At the end of this course, students will be able to

1. Identify the different types of network devices and their functions within a network.
2. Understand and build the skills of sub-netting and routing mechanisms.
3. Understand basic protocols of computer networks, and how they can be used to assist in network design and implementation.

List of Assignments:

1. Study of Networking Commands (Ping, Tracert, TELNET, nslookup, netstat, ARP, RARP) and Network Configuration Files.
2. Linux Network Configuration.
 - a. Configuring NIC's IP Address.
 - b. Determining IP Address and MAC Address using if-config command.
 - c. Changing IP Address using if-config.
 - d. Static IP Address and Configuration by Editing.
 - e. Determining IP Address using DHCP.
 - f. Configuring Hostname in /etc/hosts file.
3. Design TCP iterative Client and Server application to reverse the given input sentence.
4. Design a TCP concurrent Server to convert a given text into upper case using multiplexing system call "select".
5. Design UDP Client Server to transfer a file.
6. Configure a DHCP Server to serve contiguous IP addresses to a pool of

four IP devices with a default gateway and a default DNS address. Integrate the DHCP server with a BOOTP demon to automatically serve Windows and Linux OS Binaries based on client MAC address.

- a. Configure DNS: Make a caching DNS client, and a DNS Proxy; implement reverse DNS and forward DNS, using TCP dump/Wireshark characterise traffic when the DNS server is up and when it is down.
7. Configure a mail server for IMAP/POP protocols and write a simple SMTP client in C/C++/Java client to send and receive mails.
8. Configure FTP Server on a Linux/Windows machine using a FTP client/SFTP client characterise file transfer rate for a cluster of small files 100k each and a video file of 700mb. Use a TFTP client and repeat the experiment.
9. Signaling and QoS of labeled paths using RSVP in MPLS.
10. Find shortest paths through provider network for RSVP and BGP.
11. Understand configuration, forwarding tables, and debugging of MPLS.

CM27C: Cyber Security

Course Type:	Value Added	Semester:	II Sem	Credits :	3
Course Code:	CM27C	Theory :	2 Hrs/ Week	Practical :	2Hrs/ Week
		Assessment:	Lab reports and written exams	Internal Continuous Assessment:	100 M

Course Overview / Description: Cyber security is perhaps the most important topic in today environment. Demand for cyber security professionals has exploded, in the private and public sectors alike. Student can learn how to defend information systems from cyber attacks, how to recover compromised systems, how to architect secure systems and so much more. This course is focuses on the models, tools, and techniques for enforcement of security with some emphasis on the use of cryptography. Students will learn security from multiple perspectives.

Course Learning Objectives:

1. To prepare students with the technical knowledge and skills needed to protect and defend computer systems and networks from cyber security attacks.
2. To develop graduates that can plan, implement, and monitor cyber security mechanisms to help ensure the protection of information

technology assets.

3. To develop graduates that can identify, analyze, and remediate computer security breaches.

Course Outcomes: After completing this course student will be:

1. Be able to understand the basic terminologies related to cyber security and current cyber security threat landscape. They will also develop understanding about the Cyber warfare and necessity to strengthen the cyber security of end user machine, critical IT and national critical infrastructure.
2. Have complete understanding of the cyber attacks that target computers, mobiles and persons. They will also develop understanding about the type and nature of cyber crimes and as to how report these crimes through the prescribed legal and Government channels.
3. Be able to understand the legal framework that exist in India for cyber crimes and penalties and punishments for such crimes, It will also expose students to limitations of existing IT Act, 2000 legal framework that is followed in other countries and legal and ethical aspects related to new technologies.
4. Understand the aspects related to personal data privacy and security. They will also get insight into the Data Protection Bill, 2019 and data privacy and security issues related to Social media platforms.
5. Understand the main components of cyber security plan. They will also get insights into risk-based assessment, requirement of security controls and need for cyber security audit and compliance.

Unit- I Overview of Cyber security

Cyber security increasing threat landscape, Cyber security terminologies- Cyberspace, attack, attack vector, attack surface, threat, risk, vulnerability, exploit, exploitation, hacker., Non-state actors, Cyber terrorism, Protection of end user machine, Critical IT and National Critical Infrastructure, Cyber warfare, Case Studies.

Unit-II Cyber Crimes

Cyber crimes targeting Computer systems and Mobiles- data diddling attacks, spyware, logic bombs, DoS, DDoS, APTs, virus, Trojans, ransomware, data breach., Online scams and frauds- email scams, Phishing, Vishing, Smishing, Online job fraud, Online sextortion, Debit/ credit card fraud, Online payment fraud, Cyber bullying, website defacement, Cyber squatting, Pharming, Cyber espionage, Cryptojacking, Darknet- illegal trades, drug trafficking, human trafficking., Social Media Scams & Frauds- impersonation, identity theft, job scams, misinformation, fake news cyber crime against persons - cyber grooming, child pornography, cyber stalking., Social Engineering attacks, Cyber Police stations, Crime reporting procedure, Case studies.

Unit-III Cyber Law

Cyber crime and legal landscape around the world, IT Act, 2000 and its amendments, Limitations of IT Act, 2000, Cyber crime and punishments, Cyber Laws and Legal and ethical aspects related to new technologies-

AI/ML, IoT, Blockchain, Darknet and Social media, Cyber Laws of other countries, Case Studies.

Unit-IV Data Privacy and Data Security

Defining data, meta-data, big data, non-personal data. Data protection, Data privacy and data security, Personal Data Protection Bill and its compliance, Data protection principles, Big data security issues and challenges, Data protection regulations of other countries- General Data Protection Regulations(GDPR), 2016 Personal Information Protection and Electronic Documents Act (PIPEDA), Social media- data privacy and security issues.

Unit-V Cyber security Management, Compliance and Governance:

Cyber security Plan- cyber security policy, cyber crises management plan, Business continuity, Risk assessment, Types of security controls and their goals, Cyber security audit and compliance, National cyber security policy and strategy.

Practical Work

The practical list has been suggested for the applicable modules; however, the faculty may expand the list as per the syllabus content duly taking into consideration the emerging nature of cyber threats and incumbent protective measures to guard against such threats.

Practical list of Experiments:

1. Platforms for reporting cyber crimes.
2. Checklist for reporting cyber crimes online.
3. Setting privacy settings on social media platforms.
4. Do's and Don'ts for posting content on Social media platforms.
5. Registering complaints on a Social media platform.
6. Prepare password policy for computer and mobile device.
7. List out security controls for computer and implement technical security controls in the personal computer.
8. List out security controls for mobile phone and implement technical security controls in the personal mobile phone.
9. Log into computer system as an administrator and check the security policies in the system.

Text Books/ References:

1. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives by Sumit Belapure and Nina Godbole, Wiley India Pvt. Ltd.
2. Information Warfare and Security by Dorothy F. Denning, Addison Wesley.
3. Security in the Digital Age: Social Media Security Threats and Vulnerabilities by Henry A. Oliver, Create Space Independent Publishing Platform.
4. Data Privacy Principles and Practice by Natraj Venkataramanan and Ashwin Shriram, CRC Press.

5. Information Security Governance, Guidance for Information Security Managers by W. KragBrothy, 1st Edition, Wiley Publication.
6. Auditing IT Infrastructures for Compliance By Martin Weiss, Michael G. Solomon, 2nd Edition, Jones Bartlett Learning.

CM28M:Mini Project With Seminar

Instruction Hours/week : 4(P)

Credits : 2

Sessional Marks : 100

Semester-end Examination :-

Course Outcomes:

At the end of this course, students will be able to

1. Understand of contemporary / emerging technology for various processes and systems.
2. Share knowledge effectively in oral and written form and formulate documents.

Syllabus Contents:

The students are required to search / gather the material / information on a specific a topic comprehend it and present / discuss in the class.

Semester III & IV

(Dissertation) Dissertation Phase – I, Phase – II and Viva-Voce

Instruction Hours/week : 24+20(P)

Credits : 12+16

Sessional Marks : 40+40 Semester-end Examination : 60+60

Course Outcomes:

At the end of this course, students will be able to

1. Ability to synthesize knowledge and skills previously gained and applied to an in-depth study and execution of new technical problem.
2. Capable to select from different methodologies, methods and forms of analysis to produce a suitable research design, and justify their design.
3. Ability to present the findings of their technical solution in a written report.
4. Presenting the work in International/ National conference or reputed journals.

Syllabus Contents:

The dissertation / project topic should be selected / chosen to ensure the satisfaction of the urgent need to establish a direct link between education, national development and productivity and thus reduce the gap between the world of work and the world of study. The dissertation should have the

following

- Relevance to social needs of society
- Relevance to value addition to existing facilities in the institute
- Relevance to industry need
- Problems of national importance
- Research and development in various domain

The student should complete the following:

- Literature survey Problem Definition
- Motivation for study and Objectives
- Preliminary design / feasibility / modular approaches
- Implementation and Verification
- Report and presentation

The dissertation stage II is based on a report prepared by the students on dissertation allotted to them. It may be based on:

- Experimental verification / Proof of concept.
- Design, fabrication, testing of Communication System.
- The viva-voce examination will be based on the above report and work.

Guidelines for Dissertation Phase – I and II at M. Tech. (Electronics):

- As per the AICTE directives, the dissertation is a yearlong activity, to be carried out and evaluated in two phases i.e. Phase – I: July to December and Phase – II: January to June.
- The dissertation may be carried out preferably in-house i.e. department's laboratories and centers OR in industry allotted through department's T & P coordinator.
- After multiple interactions with guide and based on comprehensive literature survey, the student shall identify the domain and define dissertation objectives. The referred literature should preferably include IEEE/IET/IETE/Springer/Science Direct/ACM journals in the areas of Computing and Processing (Hardware and Software), Circuits-Devices and Systems, Communication-Networking and Security, Robotics and Control Systems, Signal Processing and Analysis and any other related domain. In case of Industry sponsored projects, the relevant application notes, while papers, product catalogues should be referred and reported.
- Student is expected to detail out specifications, methodology, resources required, critical issues involved in design and implementation and phase wise work distribution, and submit the proposal within a month from the date of registration.
- Phase – I deliverables: A document report comprising of summary of

literature survey, detailed objectives, project specifications, paper and/or computer aided design, proof of concept/functionality, part results, A record of continuous progress.

- Phase – I evaluation: A committee comprising of guides of respective specialization shall assess the progress/performance of the student based on report, presentation and Q & A. In case of unsatisfactory performance, committee may recommend repeating the Phase-I work.
- During phase – II, student is expected to exert on design, development and testing of the proposed work as per the schedule. Accomplished results/contributions/innovations should be published in terms of research papers in reputed journals and reviewed focused conferences OR IP/Patents.
- Phase – II deliverables: A dissertation report as per the specified format, developed system in the form of hardware and/or software, a record of continuous progress.
- Phase – II evaluation: Guide along with appointed external examiner shall assess the
- progress/performance of the student based on report, presentation and Q & A. In case of unsatisfactory performance, committee may recommend for extension or repeating the work.
